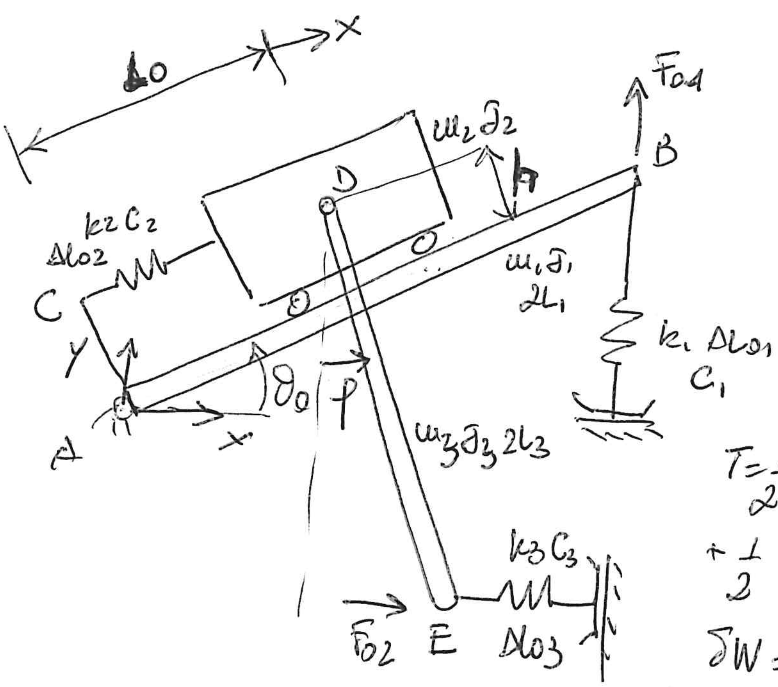


$$\underline{x} = \begin{Bmatrix} \theta \\ x \\ \varphi \end{Bmatrix}$$

$$T = \frac{1}{2} m_1 v_{G_1}^2 + \frac{1}{2} J_1 \dot{\theta}^2 + \frac{1}{2} m_2 \dot{x}_D^2 + \frac{1}{2} m_2 \dot{y}_D^2 + \frac{1}{2} J_2 \dot{\theta}^2 + \frac{1}{2} m_3 \dot{x}_{G_3}^2 + \frac{1}{2} m_3 \dot{y}_{G_3}^2 + \frac{1}{2} J_3 \dot{\varphi}^2$$

$$\delta W = F_{01} e^{i\omega t} \delta y_B + F_{02} e^{i\omega t} \delta x_E$$

$$x_D = (l_0 + x)\cos\theta - h \operatorname{sen}\theta$$
$$y_D = (l_0 + x)\operatorname{sen}\theta + h\cos\theta$$
$$x_{G3} = (l_0 + x)\cos\theta - h\operatorname{sen}\theta + L_3 \operatorname{sen}\varphi$$
$$y_{G3} = (l_0 + x)\operatorname{sen}\theta + h\cos\theta - L_3\cos\varphi$$
$$y_B = 2L_1 \operatorname{sen}\theta$$
$$x_E = (l_0 + x)\cos\theta - h\operatorname{sen}\theta + 2L_3\operatorname{sen}\varphi$$

$$\Delta l_1 = y_B - y_{B0}$$
$$\Delta l_2 = x$$
$$\Delta l_3 = -x_E + x_{E0}$$

$$\frac{\partial x_D}{\partial\theta} = -(l_0+x)\operatorname{sen}\theta - h\cos\theta \; ; \quad \frac{\partial x_D}{\partial x} = \cos\theta \; ; \quad \frac{\partial x_D}{\partial\varphi} = 0$$

$$\frac{\partial y_D}{\partial\theta} = (l_0+x)\cos\theta - h\operatorname{sen}\theta \; ; \quad \frac{\partial y_D}{\partial x} = \operatorname{sen}\theta \; ; \quad \frac{\partial y_D}{\partial\varphi} = 0$$

$$\frac{\partial x_{G3}}{\partial\theta} = -(l_0+x)\operatorname{sen}\theta - h\cos\theta \; ; \quad \frac{\partial x_{G3}}{\partial x} = \cos\theta \; ; \quad \frac{\partial x_{G3}}{\partial\varphi} = L_3\cos\varphi$$

$$\frac{\partial y_{G3}}{\partial\theta} = (l_0+x)\cos\theta - h\operatorname{sen}\theta \; ; \quad \frac{\partial y_{G3}}{\partial x} = \operatorname{sen}\theta \; ; \quad \frac{\partial y_{G3}}{\partial\varphi} = L_3\operatorname{sen}\varphi$$

$$\frac{\partial y_B}{\partial\theta} = 2L_1\cos\theta \; ; \quad \frac{\partial y_B}{\partial x} = \frac{\partial y_B}{\partial\varphi} = 0$$

$$\frac{\partial x_E}{\partial\theta} = -(l_0+x)\operatorname{sen}\theta - h\cos\theta \; ; \quad \frac{\partial x_E}{\partial x} = \cos\theta \; ; \quad \frac{\partial x_E}{\partial\varphi} = 2L_3\cos\varphi$$

$$
\begin{array}{c|c}
\begin{array}{ccc} \vartheta & x & \varphi \end{array} & \\
\Lambda_{cu} = \begin{bmatrix}
L_1 & 0 & 0 \\
1 & 0 & 0 \\
-L_0 \operatorname{sen}\vartheta_0 - h\cos\vartheta_0 & \cos\vartheta_0 & 0 \\
L_0\cos\vartheta_0 - h \operatorname{sen}\vartheta_0 & \operatorname{sen}\vartheta_0 & 0 \\
1 & 0 & 0 \\
-L_0 \operatorname{sen}\vartheta_0 - h\cos\vartheta_0 & \cos\vartheta_0 & L_3\cos\varphi_0 \\
L_0\cos\vartheta_0 - h \operatorname{sen}\vartheta_0 & \operatorname{sen}\vartheta_0 & L_3 \operatorname{sen}\varphi_0 \\
0 & 0 & 1
\end{bmatrix} &
\begin{array}{c} v_{q1} \\ \vartheta \\ x_D \\ y_D \\ \vartheta \\ x_{q3} \\ y_{q3} \\ \varphi \end{array}
\end{array}
$$

$$
\Lambda_{cr} = \begin{bmatrix}
2L_1\cos\vartheta_0 & 0 & 0 \\
0 & 1 & 0 \\
L_0 \operatorname{sen}\vartheta_0 + h\cos\vartheta_0 & -\cos\vartheta_0 & -2L_3\cos\varphi_0
\end{bmatrix}
$$

$$
\Lambda_{cQ} = \begin{bmatrix}
2L_1\cos\vartheta_0 & 0 & 0 \\
-L_0 \operatorname{sen}\vartheta_0 - h\cos\vartheta_0 & \cos\vartheta_0 & 2L_3\cos\varphi_0
\end{bmatrix}
\begin{array}{c} F_{01} \\ F_{02} \end{array}
$$

$$
h_{q1} = L_1 \operatorname{sen}\vartheta \rightarrow \frac{\partial^2 h_{q1}}{\partial\vartheta^2} = -L_1 \operatorname{sen}\vartheta \rightarrow [k_{q1}] = m_1 g \begin{bmatrix} -L_1 \operatorname{sen}\vartheta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}
$$

$$
h_{q2} = y_D \rightarrow \frac{\partial^2 y_D}{\partial\vartheta^2} = -(L_0 + x)\operatorname{sen}\vartheta - h\cos\vartheta
$$

$$
\frac{\partial^2 y_D}{\partial\vartheta\partial x} = \cos\vartheta \; ; \; \frac{\partial^2 y_D}{\partial x^2} = 0 \qquad [k_{q2}] = m_2 g \begin{bmatrix} (-L_0 \operatorname{sen}\vartheta_0 - h\cos\vartheta_0) & \cos\vartheta_0 & 0 \\ \cos\vartheta_0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}
$$

$$
h_{q3} = y_{q3} \rightarrow \frac{\partial^2 h_{q3}}{\partial\vartheta^2} = -(L_0 + x)\operatorname{sen}\vartheta - h\cos\vartheta
$$

$$
\frac{\partial^2 h_{q3}}{\partial\vartheta\partial x} = \cos\vartheta \; ; \; \frac{\partial^2 h_{q3}}{\partial x^2} = 0 \; ; \; \frac{\partial^2 h_{q3}}{\partial x\partial\varphi} = 0
$$

$$
\frac{\partial^2 h_{q3}}{\partial\varphi^2} = L_3\cos\varphi \; ; \; \frac{\partial^2 h_{q3}}{\partial\vartheta\partial\varphi} = 0 \qquad [k_{q3}] = m_3 g \begin{bmatrix} (-L_0 \operatorname{sen}\vartheta_0 - h\cos\vartheta_0) & \cos\vartheta_0 & 0 \\ \cos\vartheta_0 & 0 & 0 \\ 0 & 0 & L_3\cos\varphi_0 \end{bmatrix}
$$

$$\frac{\partial^2 \Delta L_1}{\partial \theta^2} = -2L_1 \operatorname{sen}\theta$$

$$[k_{EL,II\,1}] = k_1 \Delta l_{01} \begin{bmatrix} -2L_1 \operatorname{sen}\theta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta L_3}{\partial \theta^2} = (L_0 + x)\cos\theta - h \operatorname{sen}\theta$$

$$\frac{\partial^2 \Delta L_3}{\partial \theta \partial x} = + \operatorname{sen}\theta \;;\; \frac{\partial^2 \Delta L_3}{\partial x^2} = 0 \;;\; \frac{\partial^2 \Delta L_3}{\partial x \partial \varphi} = 0$$

$$\frac{\partial^2 \Delta L_3}{\partial \theta \partial \varphi} = 0 \;;\; \frac{\partial^2 \Delta L_3}{\partial \varphi^2} = 2L_3 \operatorname{sen}\varphi$$

$$[k_{EL,II\,3}] = k_3 \Delta l_{03} \begin{bmatrix} L_0\cos\theta_0 - h\operatorname{sen}\theta_0 & \operatorname{sen}\theta_0 & 0 \\ \operatorname{sen}\theta_0 & 0 & 0 \\ 0 & 0 & 2L_3 \operatorname{sen}\varphi_0 \end{bmatrix}$$

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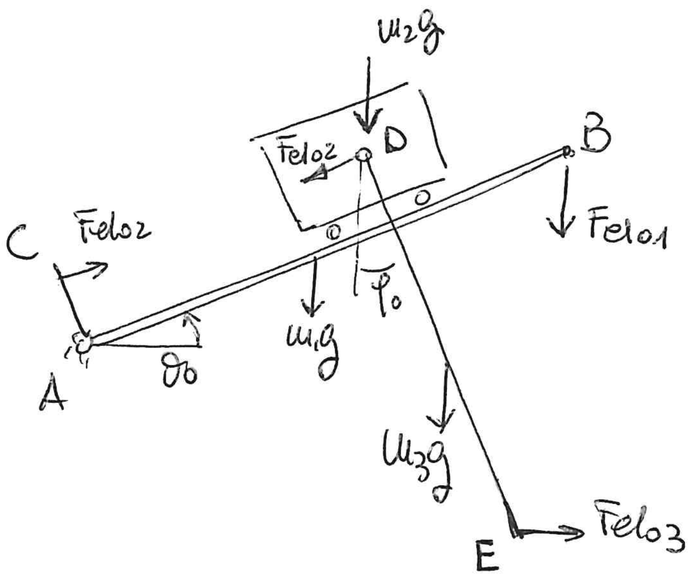



$$\sum M_D^{DE} = 0 : \quad F_{elo3}\, 2L_3 \cos\varphi_0 - m_3 g\, L_3 \operatorname{sen}\varphi_0 = 0$$

$$\boxed{F_{elo3} = \frac{m_3 g}{2} \operatorname{tg}\varphi_0}$$

$$\sum F_x^{m_2+m_3} = 0 :$$

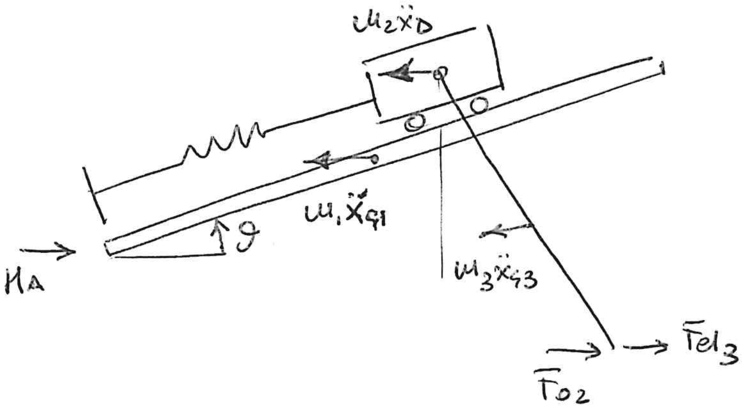
$$F_{elo3}\cos\theta_0 - m_3 g \operatorname{sen}\theta_0 - F_{elo2} + \\ - m_2 g \operatorname{sen}\theta_0 = 0$$

$$\boxed{F_{elo2} = -(m_2 + m_3) g \operatorname{sen}\theta_0 + F_{elo3}\cos\theta_0}$$

$$\sum M_A = 0 :$$

$$-m_1 g L_1 \cos\theta_0 - F_{elo1}\, 2L_1 \cos\theta_0 + \\ - m_2 g (L_0\cos\theta_0 - h \operatorname{sen}\theta_0) + \\ - m_3 g (L_0 \cos\theta_0 - h\operatorname{sen}\theta_0 + L_3 \operatorname{sen}\varphi_0) + \\ F_{elo3} (L_0 \operatorname{sen}\theta_0 + h\cos\theta_0 - 2L_3 \cos\varphi_0) = 0$$

$$\boxed{F_{elo1} = \frac{-m_1 g L_1 \cos\theta_0 - m_2 g (L_0\cos\theta_0 - h\operatorname{sen}\theta_0) - m_3 g (L_0\cos\theta_0 - h\operatorname{sen}\theta_0 + L_3 \operatorname{sen}\varphi_0) - F_{elo3}\, y_E}{2L_1 \cos\theta_0}}$$




$$H_A = m_1\ddot{x}_{G1} + m_2\ddot{x}_D + m_3\ddot{x}_{G3} - F_{D2} - F_{el3}$$